Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$$

Independent

$$\mathrm{Var}(X + Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y$$

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Worked Solutions **Expected Value and Variance of Multiple Variables**

Q001 Written

Given the probability distributions of X and Y in the source tables, find E(X+Y).

Solution

1. Find EX and EY separately, then add them.

Final answer

$EX = 35/3, EY = 9/2, so E(X + Y) = 97/6.$

Q002 Written

Three coins are tossed once; let X, Y and Z be the head indicators. Find E(X+Y+Z).

Solution

1. Each fair head indicator has expected value 1/2.

Final answer

$E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q003 **Written**

For the two source tables with X=1,2,3 and Y=4,5,6, find E(X+Y).

Solution

1. The expectation of a sum is the sum of the expectations.

Final answer

$E(X + Y) = EX + EY = 2 + 5 = 7.$

Q004 **Written**

For the two bags in the source exercise, find the expected total number of red balls drawn.

Solution

1. Use E(red drawn)=sample size times the red proportion in each bag.

Final answer

Bag A: 6/7; Bag B: 6/7; total $E = 12/7$.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find E(X+Y+Z).

Solution

1. Add the three binomial means.

Final answer

Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z.

Solution

1. Model the score as aX+bY, then apply linearity.

Final answer

$Z = 500X + 100Y, soEZ = 5001 + 1003/2 = 650.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q007 **Written**

Two 500-point coins and two 100-point coins are tossed. Find EZ.

Solution

1. The expected heads in each two-coin group is 1.

Final answer

$EZ = 5001 + 1001 = 600.$

Q008 **Written**

A die is thrown twice and the score is Z=3X+5Y. Find EZ.

Solution

1. Both die rolls have mean 7/2.

Final answer

$EZ = 37/2 + 57/2 = 28.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Solution

1. Use the expected score of each fair coin and add.

Final answer

$EZ = 350 + 25 = 175.$

Q010 Written

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

Solution

1. Expected winners are sample size times the winning proportion in each raffle.

Final answer

$EZ = 22/10 + 21/6 = 11/15.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find E(XY).

Solution

1. Check independence before multiplying expectations.

Final answer

X and Y are independent, so $E(XY) = EXEY$=3/27/2=21/4.

Q012 Written

For independent X with probabilities 2/5,2/5,1/5 at 1,2,3 and Y uniform on 2,4,6, find E(XY).

Solution

1. Compute each mean from its table, then multiply.

Final answer

$EX = 8/5, EY = 4, soE(XY) = 32/5.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find E(XY).

Solution

1. The two experiments are independent.

Final answer

$E(XY) = EXEY = 3/21 = 3/2.$

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find E(XY).

Solution

1. Replacement makes the two draws independent.

Final answer

$EX = EY = 5/2, soE(XY) = 25/4.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q015 Written

For independent X and Y with EX=1 and EY=2/7, find E(XY).

Solution

1. Use the product rule after checking independence.

Final answer

$E(XY) = EXEY = 2/7$.

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find V(X+Y) and sigma(X+Y).

Solution

1. For independent variables, add variances, then take the positive square root.

Final answer

VX=2/3, VY=3/4, so $V(X+Y) = 17/12$ and sigma=sqrt17/12.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q017 Written

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Solution

1. Each fair die has variance 35/12.

Final answer

$V(X+Y) = 35/12 + 35/12 = 35/6$; sigma=sqrt35/6.

Q018 Written

An independent coin head count X and die score Y are combined. Find V(X+Y) and sigma(X+Y).

Solution

1. Use $V(Bernoulli(1/2)) = 1/4$ and $V(die) = 35/12$.

Final answer

$V = 1/4 + 35/12 = 19/6$; sigma=sqrt19/6.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q019 **Written**

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Solution

1. Variances of independent Bernoulli counts add.

Final answer

$V = 21/4 + 31/4 = 5/4.$

Q020 **Written**

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find V(X+Y) and sigma(X+Y).

Solution

1. Use the uniform discrete variance for each scale and add.

Final answer

$V = 5/4 + 35/12 = 25/6$; sigma=5/sqrt6.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q021 Written

Three independent special dice take values 2,4,6 equally. Find E(XYZ) and V(X+Y+Z).

Solution

1. For a product multiply means; for a sum add variances.

Final answer

EX=4, VX=8/3; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 Written

Three independent dice take values 1,3,5 equally. Find E(XYZ) and V(X+Y+Z).

Solution

1. All three distributions are identical and independent.

Final answer

EX=3, VX=8/3; $E(XYZ) = 27$ and $V(sum) = 8$.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q023 **Written**

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Solution

1. Use independence for the product and variance addition for the sum.

Final answer

$E(product)$ =$7/2^3$ = 343/8; $V(sum)$ = 335/12=35/4.

Q024 **Written**

Three friends have 5, 4 and 3 independent rolls to score a six. Find E(XYZ) and V(X+Y+Z).

Solution

1. For Binomial(n,1/6), use $E = np$ and $V = np(1 - p)$.

Final answer

Means are 5/6, 2/3, 1/2; $E(product) = 5/18$ and $V(sum) = 5/3$.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q025 Written

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find E(XYZ).

Solution

1. Multiply the three expected head counts.

Final answer

$E(XYZ)$ = 23/21=3.

Q026 MCQ

E(X+Y) equals:

Solution

1. Add the two expectations.

Correct option: B

Final answer

$EX + EY$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q027 MCQ

For three variables, E(X+Y+Z) equals:

Solution

1. Extend linearity to three terms.

Correct option: B

Final answer

$EX + EY + EZ$

Q028 MCQ

The expected heads in one fair coin toss is:

Solution

1. Use the Bernoulli mean.

Correct option: B

Final answer

1/2

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q029 MCQ

For two signals with p=0.5, EX is:

Solution

1. Use np.

Correct option: B

Final answer

1

Q030 MCQ

If EX=2 and EY=5, E(X+Y) is:

Solution

1. Add the means.

Correct option: B

Final answer

7

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q031 MCQ

E(aX+bY) equals:

Solution

1. Apply linearity.

Correct option: A

Final answer

$aEX + bEY$

Q032 MCQ

The expected score of one fair 100-point coin is:

Solution

1. Half the value is expected.

Correct option: B

Final answer

50

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q033 MCQ

For a fair die, E(3X+5Y) with two rolls is:

Solution

1. Use both means 7/2.

Correct option: C

Final answer

28

Q034 MCQ

A fixed entry fee changes:

Solution

1. It is a constant term.

Correct option: B

Final answer

the mean only

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Solution

1. Use sample size times winner proportion.

Correct option: B

Final answer

2/5

Q036 MCQ

E(XY)=EXEY requires:

Solution

1. The product rule needs mutual independence.

Correct option: B

Final answer

independence

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q037 MCQ

For three fair coins, EX is:

Solution

1. Use np.

Correct option: C

Final answer

3/2

Q038 MCQ

For a fair die, EY is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q039 MCQ

With replacement, two card draws are:

Solution

1. Replacement restores the same distribution and independence.

Correct option: B

Final answer

independent

Q040 MCQ

If EX=2 and EY=3 for independent variables, E(XY) is:

Solution

1. Multiply the means.

Correct option: B

Final answer

6

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q041 MCQ

For independent X,Y, V(X+Y) equals:

Solution

1. Add independent variances.

Correct option: B

Final answer

$VX + VY$

Q042 MCQ

The variance of a fair die is:

Solution

1. Use the standard die variance.

Correct option: B

Final answer

35/12

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q043 MCQ

Sigma(X+Y) is:

Solution

1. Take the positive root.

Correct option: B

Final answer

sqrt$V(X+Y)$

Q044 MCQ

The variance of the sum of two independent fair dice is:

Solution

1. Double 35/12.

Correct option: B

Final answer

35/6

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q045 MCQ

For independent sums, a constant shift changes:

Solution

1. Translations do not change spread.

Correct option: C

Final answer

neither spread measure

Q046 MCQ

For independent X,Y,Z, E(XYZ) equals:

Solution

1. Multiply the three means.

Correct option: B

Final answer

$EXEYEZ$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q047 MCQ

For independent X,Y,Z, V(X+Y+Z) equals:

Solution

1. Add the three variances.

Correct option: B

Final answer

$VX + VY + VZ$

Q048 MCQ

For a binomial count with n=5 and p=1/6, EX is:

Solution

1. Use np.

Correct option: B

Final answer

5/6

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q049 MCQ

For three values 2,4,6 equally likely, EX is:

Solution

1. Average the three values.

Correct option: C

Final answer

4

Q050 MCQ

For three independent fair dice, V(sum) is:

Solution

1. Add three copies of 35/12.

Correct option: C

Final answer

35/4

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 1 **Concept Quiz · Written**

If EX=4 and EY=7, find E(3X-2Y).

Solution

1. Use linearity: 34-27.

Final answer

-2

Quiz WRITTEN 2 **Concept Quiz · Written**

For independent variables with EX=2 and EY=5, find E(XY).

Solution

1. Multiply the means.

Final answer

10

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 3 Concept Quiz · Written

Two independent fair dice are thrown. Find V(X+Y) and sigma(X+Y).

Solution

1. Add the two die variances.

Final answer

V = 35/6, sigma=sqrt35/6

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have (n,p)=(2,1/2),(3,1/2),(4,1/2). Find V(X+Y+Z).

Solution

1. Add np(1-p) for the three counts.

Final answer

9/4

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz MCQ 5 Concept Quiz · MCQ

If EX=3 and EY=4, E(2X+Y) is:

Solution

1. Compute 23+4.

Correct option: B

Final answer

10

Quiz MCQ 6 Concept Quiz · MCQ

For independent variables EX=2, EY=5, E(XY) is:

Solution

1. Multiply the means.

Correct option: B

Final answer

10

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz MCQ 7 **Concept Quiz · MCQ**

For two independent die rolls, V(X+Y) is:

Solution

1. Add the two variances.

Correct option: B

Final answer

35/6

Quiz MCQ 8 **Concept Quiz · MCQ**

For three independent variables, V(X+Y+Z) is:

Solution

1. Extend variance addition.

Correct option: B

Final answer

the sum of variances

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, so E(X+Y) = 97/6.$

Q002 $E(X+Y+Z) = 1/2 + 1/2 + 1/2 = 3/2.$

Q003 $E(X+Y) = EX + EY = 2 + 5 = 7.$

Q004 Bag A: 6/7; Bag B: 6/7; total $E = 12/7$.

Q005 Each mean is $2(0.5) = 1$, so $E(X+Y+Z) = 3$.

Q006 $Z = 500X + 100Y, so EZ = 5001 + 1003/2 = 650.$

Q007 $EZ = 5001 + 1001 = 600.$

Q008 $EZ = 37/2 + 57/2 = 28.$

Q009 $EZ = 350 + 25 = 175.$

Q010 $EZ = 22/10 + 21/6 = 11/15.$

Q011 X and Y are independent, so $E(XY) = EX$EY=3/27/2=21/4.

Q012 $EX = 8/5, EY = 4, so E(XY) = 32/5.$

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

Q014 $EX = EY = 5/2, so E(XY) = 25/4.$

Q015 $E(XY) = EXEY = 2/7.$

Q016 VX=2/3, VY=3/4, so $V(X+Y) = 17/12$ and sigma=sqrt17/12.

Q017 $V(X+Y) = 35/12 + 35/12 = 35/6$; sigma=sqrt35/6.

Q018 $V = 1/4 + 35/12 = 19/6$; sigma=sqrt19/6.

Q019 $V = 21/4 + 31/4 = 5/4.$

Q020 $V = 5/4 + 35/12 = 25/6$; sigma=5/sqrt6.

Q021 EX=4, VX=8/3; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 EX=3, VX=8/3; $E(XYZ) = 27$ and $V(sum) = 8$.

Q023 $E(product)$ =7/2^3 = 343/8; $V(sum) = 335/12$=35/4.

Q024 Means are 5/6, 2/3, 1/2; $E(product) = 5/18$ and $V(sum) = 5/3$.

Q025 $E(XYZ) = 23/21$=3.

Q026 B

Q027 B

Q028 B

Q029 B

Q030 B

Q031 A

Q032 B

Q033 C

Q034 B

Q035 B

Q036 B

Q037 C

Q038 C

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6$, sigma=sqrt35/6

Quiz WRITTEN 4 9/4

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B